

Enhancing MILAGE LEARN+ with Machine Learning to improve students' performance

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Abstract—Students currently attending school were born after the year 2000 and have grown up surrounded by technology, including smartphones, tablets, the Internet, video games, and social media. Typically, conventional educational activities in schools fail to engage these students, leading many of them to struggle academically. This paper presents the strategy adopted by the free MILAGE LEARN+ platform to address these challenges. Artificial Intelligence supports learning personalization by recommending suitable activities tailored to each student's individual needs, enabling both lower-performing and higher-performing learners to enhance their academic progress. This paper investigates how Artificial Intelligence, specifically through various machine learning methods, can enrich the learning experience offered by the MILAGE LEARN+ platform. Several machine learning approaches are evaluated and analysed based on data from the platform and student outcomes collected during a Mathematics course.

Keywords—artificial intelligence, mobile learning, self-assessment, peer assessment

I. INTRODUCTION

Today's students grew up immersed in digital games and social networks ([1]), making traditional educational approaches less appealing and often ineffective. MILAGE LEARN+ is a free educational platform offering resources created collaboratively by teachers and students, tailored to the national curricula of several countries ([2], [3], [4]), including Portugal, Spain, Germany, and Cyprus. The platform supports multiple disciplines such as Mathematics, languages, Physics, and Chemistry.

The MILAGE LEARN+ platform allows teachers to provide contents structured in chapters, subchapters, worksheets with tutorials and tasks. However, the platform does not offer any guidance regarding the tasks execution order. Moreover, the teacher does not have means of analyzing individual performance data and thereafter, hindering the possibility of defining individual work plans according to performance data.

This paper explores the potential of Artificial Intelligence, particularly machine learning techniques, in customizing the

learning experience provided by the MILAGE LEARN+ platform. Customizing student learning paths aims at guiding them in their choices of activities to take or problems to solve next, in order to learn more effectively as they increase their knowledge level and improving their motivation throughout the process.

The paper is structured as follows. Section II summarizes the role of Artificial Intelligence and Machine Learning in Educational Applications. Section III provides an overview of the MILAGE LEARN+ platform. Section IV presents the case study on supervised learning based on machine learning techniques and Section V presents our conclusions and future directions.

II. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN EDUCATIONAL APPLICATIONS

Artificial Intelligence (AI) enhances routine tasks by increasing their efficiency and speed [5], [6], playing an ever-growing role in daily human life. A comprehensive study [7], based on surveys from global AI researchers over six years, indicates that AI is set to exceed human skills in several areas within the next years. These areas include translation tasks (as early as 2024), composing high school essays (2026), operating trucks (2027), writing bestselling novels (2049), and even performing surgical procedures (2053). Additionally, the researchers estimate a 50% probability of AI surpassing humans in all tasks within 45 years and fully automating human labour in approximately 120 years.

Similarly, the role of Artificial Intelligence (AI) in education is becoming clearer as traditional teaching methods evolve to better meet the changing needs and contexts of students and teachers. Educational environments are increasingly personalized and tailored to individual students ([1]). AI technology offers numerous applications that are transforming how we learn, making education more accessible through computers or smart devices, both inside and outside schools. Moreover, AI also supports teachers by simplifying the analysis of student

performance and providing data-driven recommendations to inform educational decision-making.

According to a report examining the AI market within the U.S. educational sector [8], AI technology is expected to influence education across all levels, from kindergarten through higher education, by facilitating the creation of adaptive learning resources with personalized tools that improve student experiences [9], [10]. Machine Learning (ML) and Data Mining (DM) are two interconnected AI fields increasingly utilized in education. While DM techniques uncover valuable insights by analyzing data patterns, categorizing information, and revealing relationships, ML extends these findings to anticipate future outcomes based on the identified patterns.

Both DM and ML technologies support educational settings by minimizing direct teacher-student interaction, enabling learners to engage with course content in ways closely aligned with their individual needs, thus promoting a more productive and enjoyable learning journey [5]. Personalized, adaptive learning involves guiding students toward selecting appropriate lessons or exercises, enhancing their educational effectiveness. Specifically, DM techniques help identify areas where students struggle with classroom concepts or misinterpret instructions, whereas ML can forecast optimal learning pathways by evaluating a student's characteristics and current performance. Consequently, ML plays a crucial role in supporting informed decision-making to achieve personalized learning objectives.

III. MILAGE LEARN+ PLATFORM

This section describes MILAGE LEARN+ technological architecture and interface, as well as its current limitations and enhancement opportunities through the usage of Machine Learning techniques.

A. Technological architecture and interface

MILAGE LEARN+ is a web application with a client-server architecture organized as follows:

- MILAGE LEARN+ server is a Python application that allows access to a relational database implemented with PostgreSQL, managing information about users, courses, problems, and solutions uploaded by teachers to be solved by students ([2]). Currently, the application is used by around 203000 validated users that include about 195000 students and 8000 teachers.
- MILAGE LEARN+ client side ([2], [3], [4]) encompasses functionalities for students and teachers available for every platform in the web, available for free.

MILAGE LEARN+ implements a pedagogical model developed to motivate students and promote an active, student-centred, and autonomous learning providing students with educational videos, tutorials, exercises/problems, and solutions, graded through auto and peer-assessment approaches. Educational resources are organized like a book with worksheets of tasks. Worksheets are organized in chapters and subchapters. Each worksheet is structured in tutorials or questions and sections. Each section has a statement, a video with the

Fig. 1. Student's solution to a problem and immediate feedback with assessment instructions.

solution, as well as assessment instructions for both auto and peer-assessment tasks.

With the MILAGE LEARN+ platform students solve problems, and they have immediate feedback that is used for self-assessment (Figure 1). After solving the problem and doing the self-assessment, the student can choose to watch an educational video to study the problem solution. In this way, the student can watch the video as many times as those needed to learn how to solve that problem. This is a step for the inclusion of low achiever students in the process of learning. MILAGE also provides three levels of problems: beginners, intermediate and advanced to accommodate the low achievers and also motivate top performers students to use the app. After completing a worksheet of tasks, the student will assess a peer. In this way, the student revises the contents when assessing the solution of a problem solved by another student.

MILAGE LEARN+ functionalities for teachers promotes the development of a sharing community of teachers and student authors, enhancing the development of personalized resources and soft skills. Teachers can follow the work and access the digital portfolio of students. In this way, the teacher can also check the self and peer assessment.

B. Enhancing MILAGE LEARN+ with Machine Learning

Currently, the guidance a student receives regarding their learning paths is pre-defined by teachers through chapter and subchapter numerations, and the level of difficulty attributed to the worksheets inside them. Consequently, the path suggested is the same for all students regardless of their abilities and learning styles. Such a one-for-all learning approach hinders MILAGE LEARN+ potential in providing more effective and

customized learning experiences. The overview in Section II shows that Artificial Intelligence, particularly Machine Learning techniques, are a promising means to overcome the current limitations of the platform. Table I summarizes two MILAGE LEARN+ limitations and potential solutions provided by ML to overcome them.

The remainder of this paper summarizes the results of a project that addresses the previous limitations. The main purpose of these enhancements through ML is to enable students and teachers seeing projections of customized learning paths based on individual historical data, providing students a learning experience based on their individual capabilities and learning styles. It will also enable professors to track and manage the progress of each student, based on projections of their performance. The large number and variety of ML techniques required in first place selecting the proper type of techniques to explore. Considering the nature of the problems to be addressed, the following assumptions were established:

- Since MILAGE LEARN+ keeps a record of students' classifications, data is already labeled. Thereafter, the project focused on supervised learning techniques.
- Considering the numerical and continuous nature of the data (classifications), the project focused on regression techniques.
- From performance data, the potential of classification models from categorical variables to define students' achievement levels is also explored.

IV. METHODOLOGY

The methodology chosen for this work was the Cross Industry Standard Process for Data Mining (CRISP-DM) ([7]). CRISP-DM, the most common methodology for data mining, analytics and data science projects, provides an adequate framework that guides all the phases encompassed by this kind of project. The life cycle of a data mining project consists of six phases. Figure 2 shows the CRISP-DM process phases.

MILAGE

This initial phase focuses on understanding the MILAGE objectives and requirements that motivate the project and

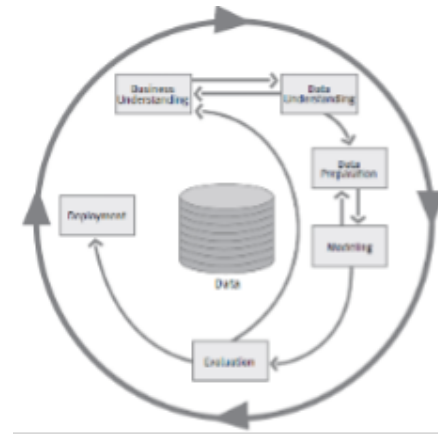


Fig. 2. CRISP-DM Phases ([7]).

translating them in terms of a data-analytic problem. A preliminary plan designed to find a solution is also included. The phase encompasses: (1) determining MILAGE objectives, (2) assessing the situation, (3) determining data mining goals and (4) producing the project plan. In the present work, this phase was implemented through several meetings with the MILAGE LEARN+ team, used to communicate the teaching and learning domains surrounding the usage of the application. Section III summarizes the results of steps (1) to (3). The meetings, which took place throughout the all project, allowed initially defining the project plan and course of actions and later, correcting them, according to the results and new information provided and discussed.

Data understanding

The data understanding phase encompassed data collection followed by data analysis, to get familiar with the data, identify data quality problems, discovering preliminary insights or formulating hypotheses. In this work, a detailed analysis of the entity-relationship diagram, was conducted to understand all the information stored in the MILAGE LEARN+ database. The relationship between each data item and the application functionalities were also identified.

After understanding the structural properties of data, an analysis of relevant properties of the data was conducted. The dataset corresponds to data from students of year 9 from the mathematics discipline. The selected academic year includes about 16718 students in the mathematics discipline that started in September and ended in June. Altogether, these students submitted and self-assessed 270124 solutions. The total dataset includes a total of about 1400 tutorials, 32000 questions, each one with a variable number of sections for a total of 55000 sections. Questions are distributed to students grouped in worksheets. There are about 11500 worksheets in total classified in 3 levels of difficulty.

This analysis investigated the answers to sections that had been assessed by the student, a peer, and the professor. A total of 1237 sections with 231288 answers by 11096 students satisfied these criteria. In this analysis, the average

TABLE I.
LIMITATIONS AND POTENTIAL SOLUTIONS THROUGH MACHINE
LEARNING

Limitation	Potential solution through ML
Teacher content is standard for all students and is shown in linear fashion, without any content recommendation based on student previous results.	Usage of a ML technique capable of recommending students contents based on projected results, and performance classification from historical data from past exercises.
MILAGE LEARN+ does not show teachers projections of students' performance from their classification record.	Usage of a ML technique capable of predicting a student future performance based on the historical performance of the student.

number of sections answered by students was calculated, the assessment values registered, and the performance achieved. These calculations allowed us to identify that the average performance level of all students was between 60% and 72%, and that few students exhibited a performance greater than 70%, corroborating the need of devising means to improve these performance levels.

Finally, a verification of the data quality, allowed detecting several anomalies:

- (a) Data inconsistency. For example, it was found that when the state of an item at a given hierarchical level was changed, the new state was always not propagated to the items in the lower levels, or some classification values exceeding the maximum predefined for the question. The latter example is illustrated in Table II and Table III .
- (b) Lack of important attributes. For example, there was no attribute to register the professor evaluation.
- (c) Lack of balance. For example, when auditing the total classification of the student Id25509 (see Table IV) from the total points of 8461, 5990 were given through auto-assessments and the remaining through peer assessments. Also, no assessment records from users with teachers' profile were found.

TABLE II.
USER ANSWER LOG

id_user_answer	id_section	points	id_user_avaliator
336	703	35	390

TABLE III.
QUESTION METADATA

questions_idquestions	id_section	points	is_multiple_question
312	703	25	n

TABLE IV.
POINTS AUDIT TO STUDENT ID 25509

Student record:	23763	23765	23772	24151	25509	30720	points
25509	731	517	70	197	5990	956	8461

Data preparation

Chapman ([7]) describes the data preparation phase as follows:

The data preparation phase covers all activities to construct the final dataset (data that will be fed into the modeling tool(s)) from the initial raw data. Data preparation tasks are likely to be performed multiple times and not in any prescribed order. Tasks include table, record and attribute selection as well as transformation and cleaning of data for modeling tools.

In the present work, this phase encompassed the analysis of 829709 records and the removal of records or attributes with the problems identified in the previous phase:

- Time-stamp attribute with year
- Records where user register and creation data are blank
- Attribute "is_multiple_question" is empty since for the purpose of this work only questions with multiple selection answers will be considered
- Records without assessment events
- Since many of the quality problems were related to the initial usage of the application, all records before september were removed. The answers to inactive worksheets, questions or sections were included since they were part of the students' historical record.

After concluding data cleaning process, the initial number of records was reduced to 648234 corresponding to 293 different chapter, 746 topics with its related chapters, sub-chapters worksheets, questions and sections belonging to 24414 students from 820 schools. The next step involved feature engineering tasks, with several calculations to derive relevant new information from existing attributes. First, to reduce the number of records, data was aggregated at the level of the topic. In this aggregation, section, question, and worksheet answers were summarized at the sub-chapter level. Some derived attributes included the total number of sections answered by a student for a given sub-chapter, total points achieved by a student for a given sub-chapter, and the average points achieved by all students in each sub-chapter. This aggregation resulted in a further reduction from 648235 records to 77975 records.

Modeling

Typically, there are several techniques for the same data mining problem type. Therefore, in this phase, various modeling techniques are selected and applied, and their parameters are gradually adjusted to obtain optimal performance values. Since some techniques have specific requirements regarding the nature and type of data available, stepping back to the data preparation phase is often necessary. The aim of this phase was to explore the possibility of providing content recommendations. Figure 3 depicts the steps of the Project phases, including the modeling phase. The modeling tools used in this work were Microsoft Power BI with AutoML, MATLAB R2021b with App Toolbox Machine Learning, and RapidMiner 9.9 with Auto Model.

Power BI with AutoML (automated ML) allows an automatic process encompassing data cleaning, correction, construction and aggregation. First, it is defined the dataset to which the ML pipeline will be applied, and the attributed to be predicted (Y attribute) is defined. The Y attribute selected in this work was the total number of points and the classification achieved by a given student in a specific sub-chapter. The former variable indicates the performance (numeric value) of each student in each sub-chapter and the latter aggregates the results in classification groups. The goal is two-fold. First, to create a model capable of predicting the performance that a given student will have in a particular sub-chapter. Second, to predict the results in terms of the defined classification groups. As mentioned in Section III, since the performance

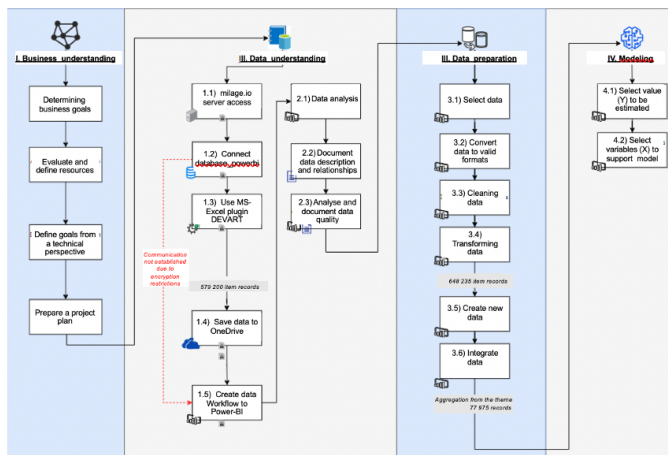


Fig. 3. Project phases and steps

attribute is numeric the recommended model is a regression model. In predicting a classification group, it is recommended a classification model. Power BI does indeed recommend these models for each predicted attribute.

After selecting the model, it is necessary to choose the independent attributes (X) that will enable to define a predictive function with the greatest generalization capability. A correlation analysis facilitates the attribute selection task. Once X attributes are selected, it is necessary to define the time limit for training the model. Power BI performs the data separation in training and testing data automatically. Such separation is performed randomly where 80% of data is used to train the model and the remaining 20% to test it.

MATLAB was used exclusively in model training and testing since the previous phases were already performed with Power BI. After importing the dataset, it was only necessary to identify Y and X attributes, as well as their datatypes. Data separation in MATLAB is performed activating the Cross Validation option. MATLAB was then configured to test several models, which allows testing all available models. The Auto Model resource included with RapidMiner also facilitates the ML process. This resource also allowed us to assess and prepare the MILAGE LEARN+ dataset, as well as creating ML models from the models suggested by RapidMiner. After selecting the model, the user proceeds to select the Y attribute for the data sample (performance and classification group). For the performance value, being numeric, Rapid Miner also suggested a regression model. After that, the user is asked to select the X attributes.

Evaluation

At this stage in the project, you have built a model (or models) that appears to have high quality from a data analysis perspective. Before proceeding to final deployment of the model, it is important to thoroughly evaluate the model and review the steps executed to construct the model to be certain it properly achieves the MILAGE objectives. Assessing model quality entails using metrics allowing us to calculate and

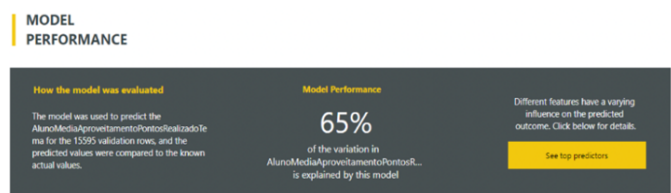


Fig. 4. Model performance with Power BI.

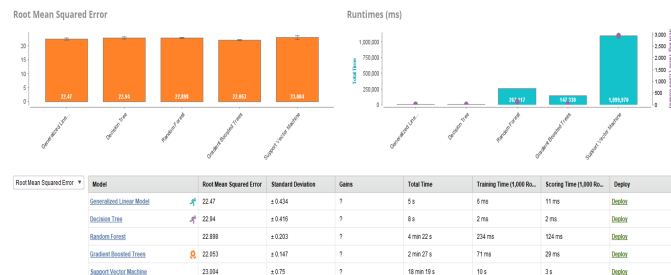


Fig. 5. Results with Power BI.

analyze the differences between estimated and actual values ([7]). In this sense, the regression models were assessed using the Root Mean Square Error (RMSE), calculated from the residual value i.e., the difference between the predicted value and the actual value for each data. The classification models were assessed by the PRECISION metric, which divides the sum of correct predictions by the sum of all predictions. A key objective of this phase is to determine if there is some important MILAGE issue that has not been sufficiently considered. For that purpose, a report was also elaborated with a summary of the experiments performed with its results. At the end of this phase, a decision on the use of the data mining results should be reached. As depicted in Figure 4, model results with Power BI only show the model with the best performance (65%), which was the Light GBM Model. The RSME value was calculated using the RMSE formula using the Power BI Desktop tool. The RMSE value obtained was 20.98.

Figure 5, exhibits results obtained with Rapid Miner, which shows results for 5 models and their respective RSME values. The best performance is achieved with the Gradient Boosted Trees model with a RSME of 22053. MATLAB shows results for 24 different models, with the best performance achieved by the Gaussian Process Regression (Exponential Regression), RMSE = 21.91.

In the classification task, it was aimed at predicting a student classification within 5 categories. The model recommended by Power BI for this task is the Pre-fitted Soft Voting Classifier (PRECISION = 86%). Rapid Miner recommends its Generalized Linear Model (PRECISION = 69%). MATLAB recommends the Support Vector Machine Quadratic (SVM) with a precision of 86,6%). When comparing the results from Power BI, Rapid Miner, and MATLAB, whereas the best regression model is provided by Power BI (Light GBM Model,

RMSE 20.98), the best classification model is provided by MATLAB.

Deployment

Model creation is not the last step of the project. Even if the purpose is to discover data patterns, it will be necessary to be organized and present the discoveries to the customers or final users. In the future, this project will involve the final integration of ML algorithms in the MILAGE LEARN+ platform. This work did not deploy the models explored. However, it was discussed how to implement the models to provide the content recommendations within MILAGE LEARN+, based on the project goals and technological architecture of the application. In this regard, it is noteworthy that Power BI does not allow configuring the type of model to use. Moreover, executing the model it is not possible to export the model and use it in another execution environment other than the provided by Power BI. Rapid Miner allows exporting and configuring the model to be used in other execution environments through web services. MATLAB provides an extensive model library, and it is possible to choose and configure the model to train. It is also possible to export the model and execute it on other execution platforms. Thereafter, Rapid Miner and MATLAB enable a better integration of ML with the MILAGE LEARN+ platform. The approach suggested for the implementation within MILAGE LEARN+ is through a separate ML module with access to performance and classification data and, encompassing the models to be used, on demand, by students that aim at increasing their accumulated punctuation or improving its classification group. In both cases, MILAGE LEARN+ will read the model results and will indicate to the student the sub-chapters that will enable the student to achieve the intended goal, according to the model results. It is important to note that the recommendation should filter out the themes already performed by the student.

V. CONCLUSIONS AND FUTURE WORK

This paper explores the potential of machine learning techniques, namely, regression and classification in enhancing the learning experience of MILAGE LEARN+ users with content recommendations. The aim is to improve students' performance and classification levels. The platform gives feedback to students after solving a problem, providing the solution together with the criteria for assessment for self and peer assessment. However, currently the platform provides little guidance regarding their learning path. The tests were performed using dataset with performance and classification data of Year 9 Mathematics course, using three different products; Power BI from Microsoft, Rapid Miner and MatLab. Results of regression tests show a better performance of PowerBI, and a better performance of MatLab in the classification tests. Results show that the techniques explored are promising in computing indicators of student workload, performance and progress over time. Nonetheless, for new users it is necessary to create profiles with performance information to guide students providing content recommendations as soon as

the user starts the learning process. The ultimate goal is to integrate the models within the MILAGE LEARN+ platform to allow real-time access to the student performance data. The approach proposed is to extend MILAGE LEARN+ through a data analytics component with real-time access to students' performance and classification data. In this sense, Rapid Miner and MatLab provide more means to facilitate such integration. Machine Learning techniques can also be used in assessing the platform contents and to improve the quality of the contents and in identifying the skills that need to be developed or deepened by the students. A further application of these techniques is to collect students' grades from a given school at the end of each term and use it to calibrate students' performance and progress indicators computed from the MILAGE LEARN+ database.

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